

MP6: Simple (fake) Modeling Editor: Polygonal Modeling and File Texture Mapping

Due Time: Refer to the course web-site

Objective

This programming assignment would be our first attempt at building something slightly more elaborate that supports (fake) interesting user interaction. We want to verify our understanding for MVC architecture, knowledge in indexed triangles, polygonal modeling, and, UV file texture mapping.

Approach

We will build a simple polygonal modeling editor with file texture transform control. Here is [my attempt at implementation](#) for your reference.

There are four main parts to this assignment: 1. Planar Polygonal Modeling, 2. Texture Placement Support for Planar Model, 3. General Cylinder Modeling, and 4. Camera Manipulation.

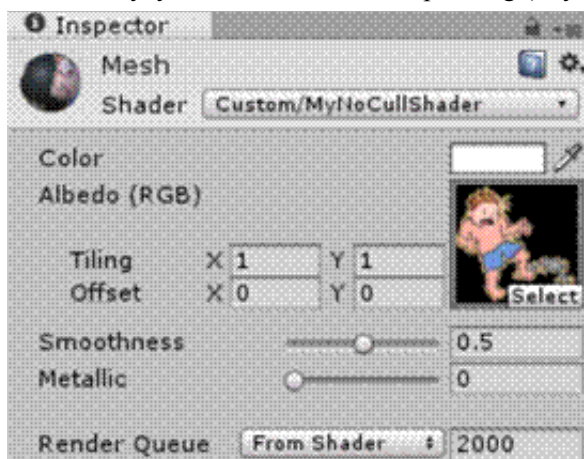
Planar Polygonal Modeling: we want to verify two main concepts: A. you understand and can implement NxN mesh based on indexed triangle; B. you understand MVC model sufficiently well such that you can implement a general translation manipulator. To help verify these, you will:

- **Basic Model:**
 - Support a slider bar that allow your user adjust the resolution of the NxN mesh. The range of the slider bar must support between 2 to 20.
 - When user adjusts the mesh resolution slider bar, you can re-initialize the mesh (thrown away user work is ok), but you must interactively show the new mesh resolution.
- **Direct User Manipulation:**
 - **Switch on vertex manipulation:** When the user type the left-control key, you will switch on vertex manipulation information.
 - **Vertex manipulation information:** User selectable spheres and normal direction at the vertex position.
 - **Selection of a vertex:** left-control + Left-Mouse-Button (LMB) click on a vertex manipulation sphere will select the vertex
 - **Selected vertex:** the sphere representing the selected vertex must be displayed in a unique color
 - **Translation manipulator:** when a vertex is selected, a translation manipulator must be displayed anchoring at the selected vertex position:
 - **Intuitive interface:** You user must be able to interact with the manipulator in an *intuitive manner* to accomplish the goal of translating the selected vertex
 - **My implementation:** in my implementation, I displayed an axis-frame-like three cylinder (with R, G, B colors) to represent the three major axis for user to click and manipulate the selected vertex X/Y/Z position. Imitating Unity, the axis being editing changes its color to Yellow.
 - **Manipulation results:** must be shown interactively at all time (with proper normal directions showing on the mesh).
- **CSS551 only (CSS451 3-member teams or Extra credit):** Support a general resolution of NxM where N may or may not equal to M.
- **Implementation note:**
 - Please use the 451NoCullShader (with Cull Off) so that we do not have to deal with backface culling in this assignment.

Texture Placement Support: To verify our understanding of transform TRS, you will support the transform manipulation of UV space on the Planar Polygonal object:

- **Specifying T/R/S:** Provide a way for your user to specify the translation, rotation, and scaling to be performed on the UV space.
 - **My implementation:** I provided a GUI object to support this operation.

- **Interactive response:** You must interactively display the results of user's TRS manipulation on the Planar Model.
- **Implementation constraint:** you must work with the [Matrix3x3 library](https://gist.github.com/anyuser/bf3b6b76937f778636771153d8ba4ff6/revisions) (Please download the zip, and unzip into your Assets/Source folder, this library is downloaded from: <https://gist.github.com/anyuser/bf3b6b76937f778636771153d8ba4ff6/revisions>) and compute transformed UV values in the CPU and not use the Unity's default texture placement functionality.
 - This is to say, you should not be manipulating (in your scripts) the Unity material Albedo-Tiling/Offset fields:



- Note: the default Unity texture placement does not support rotation, and you must!

General Cylinder (also referred to as Rotational Sweep) **Editing Support:** To verify our understanding of simple modeling with algorithms, you will support:

- **Selection of the General Cylinder:** Your user must be able to switch between General Cylinder and Planar Editor at will at any time.
- **Parameters:** your general cylinder must support these parameters
 - **Resolution:** a slider bar to allow user specify the resolution of the cylinder. This should be a number between 4 to 20.
 - **CSS551 only (CSS451 3-member teams or Extra credit):** Support a general resolution of NxM where N may or may not equal to M.
 - **Rotation degree:** a slider bar to allow user specify the degrees to sweep: min of 10, and max of 360.
 - **CSS551 only (CSS451 3-member teams or Extra credit):** When rotation is 360 degrees, make sure the normal vector at the touching vertices are reconciled such that there is no illumination gap.
- **Direct Manipulation** on the vertices along the control profile: your user must be able to select vertices along the control profile and perform direct manipulation on them just the way they manipulated the vertices on the planar object.
 - Same control workflow: left-control button to activate the selection spheres/normal, left-control+LMB to manipulate the translation manipulator object.
- **My implementation bug:** In my implementation, I did not initialize the cylinder until it is first displayed by the user. This means, if you try to change a cylinder's resolution, before it was first displayed, my system will fail. Try doing this to crash my system:
 - Start, change Cylinder Resolution slider bar, drop-down menu select cylinder to see a boring white plane. At this point the cylinder is messed up. Don't allow this to happen to your implementation!!
- **Illumination with light sources:** **CSS551 3-member teams only (extra credit for everyone else):** Support two light sources with simple diffuse illumination
 - A directional light: from the eye position parallel to current viewing direction
 - A point light: where the user has control (over 3x slider bars to control the position of the light)
 - On/Off ability (ok to map to keyboard): ability to switch each light on/off

Camera manipulation: To re-verify our understanding of how to manipulate the camera, AND, to understand how to integrate various input requirements, you will integrate camera manipulation results from MP6 into this assignment:

- You will display a **look at** position in the scene.
 - You DO NOT need to support explicit manipulation of this position (slider bar or otherwise)

- **Tumble/Track/Dolly:** you must support all these operations with the Left-ALT + LMB button much as you have in MP6.
- **Conflict:** when both the Left-ALT and Left-Control buttons are pressed, Left-ALT has priority and director manipulator should ignore mouse operations.

Hints:

- During development/debugging, ALWAYS leave the normal vectors on!
- Make sure you think through your Direct Manipulation workflow. Remember, the Direct Manipulator can be think of as a MVC unit
 - Model is the Transform (actually, the localPosition)
 - View is the three-cylinder object
 - Controller is one of the cylinders
 - **Be carefully:** the selection spheres are probably children of your Mesh (locations are probably between -1 to 1). You are going to control the sphere locations in the world (with consideration of the Mesh transform). This means, you should set the sphere's localPosition, and yet, you want to set your manipulator to the Transform.Position of the sphere (such that the Mesh's transform information is properly considered).
- Remember to BACKUP often!! I back up by duplicating the entire project folder, about every 20-30 minutes!!
 - I had to go back to previous saved versions, VERY often.
 - I have about 7-versions in my implementation, as soon as I have something interesting functioning, I make a complete copy of the project and save that in a separate folder (using the date+time as the name of the folder).
- I need to use Awake() call again, for setting the default resolution of my cylinder. My cylinder class is a subclass of the Mesh, and has a grand-total of less than 50 useful lines of code.
- **Note:** if you cannot display normal vector at all, it will be challenging to grade and you may resulted in losing a **significant percentage of your grade** because we cannot verify if your system is working properly or otherwise. You should make normal vector display as one of your highest priorities.

Credit Distribution:

1.	Planar Polygonal Modeling Support:		25%
	a. Slider bar to allow 2 to 20 vertex resolution change b. Real-time response to resolution change (my solution: when Left-Control is pressed) c. Normal information is correct at all time d. Support NxM resolution control (CSS551/Extra Credit for CSS451)	5% 10% 10% 10%	
2.	Texture Placement Support		15%
	a. Intuitive support for user to specify T R S on the transform (GUI is ok) b. Interactive response: texture changes immediately during manipulation Grader: please ensure texture rotation is working	10% 5%	
	Direct Manipulation for Vertex Translation		30%
	a. Support of Left-Control key activation of display spheres/vectors b. Showing/Hiding the manipulator for the selected vertex (sphere) c. Selected vertex has a different color d. Proper/Intuitive translation manipulation	5% 5% 5% 10% 5%	

3.	support/results e. Proper feedback during the manipulation (axis being manipulated show different color, Yellow in my case) Note: Again, without normal display, it will be very difficult to verify correctness and you will run the risk of losing significant credits.		
4.	General Cylinder Edit Support a. Able to switch editing object between Planar and Cylinder object 5% b. Slider bar to allow 4 to 20 vertex resolution change 5% c. Slider bar to allow sweep angle control between 10 to 360 degrees 10% d. Interactive display of all changes 10% e. Support NxM resolution control (CSS551/Extra Credit for CSS451) 5% f. Ensure normal is properly handled at the seam when rotation is 360 (CSS551/Extra Credit for CSS451) 10% CSS551 3-person Team: 15 - Support for Directional Light: 5 - Support for Point Light with Position Control: 5 - Support for each light on/off: 5		20%
5.	Camera manipulation support a. Proper support for Alt-LMB and Alt-Control-LMB 5% b. Camera Tumble/Track/Dolly all works fine 5%		5%
6.	Submission and others: (proper folder names, EXE folder, windowed, resizable, etc.)		5%

Bold Red fonts indicate CSS551-specific requirements. CSS451 students *do not* need to work on these (extra credits)

Green (bold) fonts: CSS551 3-person teams, extra credit for everyone else

This is a **two (or three) person team assignment** (team according to your final project team).

Extra Credit: Some ideas, DO NOT work on the following until you are completely done with the base assignment.

- Support Extrude: https://en.wikipedia.org/wiki/Polygonal_modeling with proper and meaningful UI.
- Support a *meaningful* small view camera: where the position and viewing direction of the Main Camera must be shown in the small view at all time. Support a meaningful small camera manipulation.
- Support saving/loading of the modelled object.
- Support showing/selection/editing of multiple modelled objects in the scene at one time.
- Support multiple blended layers of texturing and allow placement operation to be performed on each layer.
- Reminder:** one can only get a maximum of 20-pt extra credit from any single assignment.